Andre Arcaina

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Education

Toronto Metropolitan University (Formerly Ryerson University) Sep 2022 – Apr 2027
Bachelor of Science (Honours), Computer Science (Co-op)
Coursework: Data Structures, Algorithms, Computer Networks, Databases, Operating Systems, Big Data, Distributed Systems

Experience

Cloud Native Computing Foundation (CNCF) Aug 2026 – Present
Open Source Contributor *Markham, ON (Remote)*

- Contributed 3+ merged PRs to the `Kepler` project, a Kubernetes-based Prometheus exporter for energy consumption metrics, by fixing Helm chart bugs, improving end-to-end test coverage, and strengthening CI workflows.
- Submitted 4+ modernization fixes across the Flux CD ecosystem, improving testing and CI workflows in `flux2`, `cli-utils`, `source-controller`, `kustomize-controller`, and `helm-controller`.
- Investigated and implemented fixes in `grpc-go`, contributing to the reliability and maintainability of the gRPC ecosystem.

Environment and Climate Change Canada (ECCC) Jun 2025 – Aug 2026
Software Engineer Intern *North York, ON*

- Modernized 10+ mission-critical Java applications to Java 17 and Jakarta EE 10, reducing security risk and technical debt for national weather, maritime, and Coast Guard operations.
- Resolved 5+ production issues by investigating C++ decoder and Java service errors in Graylog, monitoring throughput and rejected products with Grafana and Prometheus.
- Built a Go end-to-end test suite integrated into Jenkins CI, automating 25 scenarios to validate a satellite decoding system.
- Refactored three C++11 satellite data services, replacing outdated error-handling code to make failures easier to catch.

Undergraduate Science Society of TMU (USSTM) Jan 2025 – Present
Backend Engineer *Toronto, ON*

- Co-owned a contract-first Go REST API with 60+ routes for a campus item-lending service with 290+ items, deployed and maintained on Amazon EC2 with Docker Compose and Caddy.
- Reduced regression risk by adding 40+ unit and integration tests for catalog, borrowing, checkout, and authorization workflows, reaching roughly 70% coverage of core application packages.
- Integrated Redis-backed Asynq workers with AWS SES to asynchronously deliver one-time login codes for the passwordless authentication flow, processing and sending 11+ emails.
- Implemented AWS S3 storage with presigned URLs so users can securely upload item and condition photos directly.

DataKinetics May 2024 – Aug 2024
Software Engineer Intern *Ottawa, ON*

- Led development of a Spring Boot REST API and dashboard that converted COBOL Copybooks to JSON, letting engineers edit Db2 records and decode mainframe data for troubleshooting.
- Built 75 Postman requests and 97 assertions validating API responses, pagination, and error handling across three Db2 tables.
- Documented the dashboard in Confluence for 12 R&D stakeholders, creating a shared reference for testing and onboarding.

Projects

Homelab Kubernetes Cluster | Linux, Virtualization, Networking, Observability, GitOps, CI/CD

- Deployed a private cloud on Proxmox VE with a three-VM K3s cluster hosting personal media, public projects, and private services.
- Built a CI/CD GitOps pipeline using GitHub Actions and Flux CD to automate tests, container builds, and Kubernetes deployments.
- Implemented observability using Prometheus, Grafana, Loki, and Grafana Alloy across 11 services, with dashboards and alert rules for diagnosing infrastructure and application failures.

Distributed Key-Value (KV) Database | Go, Raft, BadgerDB, Docker, Kubernetes (kind) GitHub

- Built a five-node replicated key-value database with linearizable HTTP/gRPC APIs and automatic leader election.
- Containerized and deployed the cluster, then chaos-tested network partitions, leader failures, abrupt node crashes, and concurrent writes to verify fault tolerance and data consistency.

Skills

Languages: Go, C++, Java, Python, JavaScript, TypeScript, SQL, YAML, Shell (Bash)
Frameworks/Data: Spring Boot, FastAPI, React.js, Next.js, gRPC, PostgreSQL, Redis, NATS
DevOps/Tools: GitHub Actions, Jenkins, GitOps (Flux CD), Prometheus, Grafana, Loki, Graylog, Postman, Git, GNU Make
Infrastructure: Kubernetes, Docker, Helm, Linux, Traefik, Tailscale, Cloudflare, AWS (EC2, S3, SES), GCP (Cloud Run, Cloud Storage)